

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 6-1-2025	Meeting No.: 4
Meeting Mode: Face-to-Face	
Project ID: 860	Project Type: Research-based
Project Title : Autonomous Driving Agent Training Using Deep Reinforcement Learning in CARLA Simulator	
Student ID : 242UC243LS	Student Name: NG JUN XUAN
Student Programme and Specialisation: Data Science	
Supervisor Name: Dr. Mohammad Shadab Khan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

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1. WORK DONE

Tasks: Background Study or Literature Review / Requirement Analysis or Theoretical Framework / Design or Research Methodology

Details (in point form):

- 2.1. Introduction
- 2.2. Systematic Literature Review Methodology
- 2.3. Specific Applications of Transformer-RL in Autonomous Driving
- 2.4. Comparative Applications of RL Algorithms in Autonomous Driving Decision-Making
- 2.5. Definition and Research of High-Risk Autonomous Driving Scenarios
- 3.1. Introduction
- 3.2. Comparative Analysis of PPO and SAC
- 3.3. Comparative Analysis of SAC and DDPG/TD3
- 3.4. Comparative Analysis of Merging and Intersection Scenarios DDPG/TD3 and PPO
- 3.5. Comparative Analysis of Merging and Intersection Scenarios PPO, SAC and DDPG/TD3
- 3.6. Analysis summary

2. WORK TO BE DONE

Tasks: Literature Review / Requirement Analysis or Theoretical Framework / Proof of Concept

Details (in point form):

- Chapter 4: Scenario risk analysis
- Chapter 5: Analysis of Transformer Variants

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

Problem:

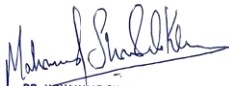
There is a lack of a unified standard for measuring the merits of algorithms. Because different papers use different simulation environments, it is difficult to directly quantify and compare the performance of PPO, SAC, and DDPG and to accurately rank them.

Solution:

The solution is to use multi-dimensional metrics (such as collision rate and success rate) for horizontal comparisons and to focus on papers that directly compare multiple algorithms under the same experimental conditions.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Satisfactory Progress.



DR. MOHAMMAD SHADAB KHAN
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Supervisor's Signature



Student's Signature

Co-Supervisor's Signature
(if applicable)

Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.